

Class 3: XG Architecture Basics

RAN, Core, and O-RAN

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CIS 4930/5930 Future Edge Networks

<https://xinliulab.github.io/FSU-CIS4930-CIS5930-Future-Edge-Networks/>

What does the icon actually tell you?



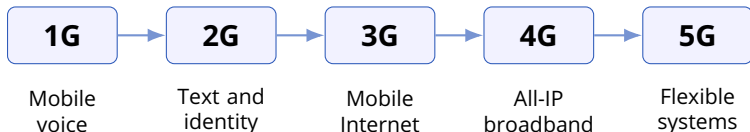
Ericsson active 5G antennas

Turn off Wi-Fi for ten seconds.

What appears: LTE, 5G, 5G+, or 5G UW?

Does that icon describe the radio, the core, or the whole network?

Each generation made a new habit feel normal



The app was visible. The architecture underneath was not.

The first mobile call was basically a flex



Motorola DynaTAC 8000X

Martin Cooper called his rival.

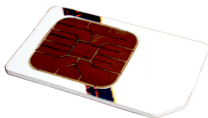
- The prototype call happened in 1973.
- Commercial analog cellular service followed in the 1980s.
- The service ordinary users noticed was mobile voice.

1G moved the telephone away from the wall.

The SIM card is the part that knows you



Nokia 3310



GSM SIM card

2G went digital.

- Digital voice and SMS
- SIM-based subscriber identity in GSM
- Mass-market roaming and feature phones

Which is “you”: the phone, the SIM, or the number?



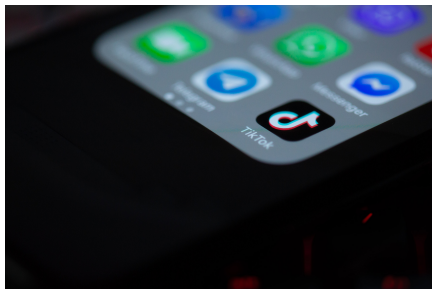
Nokia 6680 commercial (YouTube)

Data became central.

- Web, email, multimedia
- UMTS/WCDMA radio access
- Packet data grew beside legacy voice



4G made buffering feel like a personal insult



TikTok app on phone

Broadband became the default.

- Streaming and short video
- Maps, ride sharing, live apps
- Voice moved toward IP with VoLTE

4G did not invent video. It made mobile video routine.

The 5G icon can lie by omission



Trying AT&T's "5G E" (YouTube)

"Why is LTE faster than 5G here?"

Paraphrased user question, 2026

- The icon mainly reflects radio access and branding.
- Coverage, spectrum, congestion, and deployment mode matter.
- A 5G radio can still depend on a 4G core.

You press Call. What happens next?



UMTS video call

Your phone has to find a tower.

The network has to recognize you.

Then it has to find the other end.

Which part of the network
does each job?

Four worlds are hiding behind one tap



RAN

Turns radio waves into managed links.

Core

Remembers who you are, where you are, and what you may do.

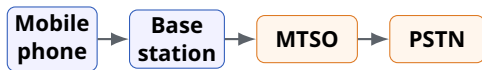
Conceptual model based on 3GPP system architecture

The network still has the same six jobs

- 1 Find a cell
- 2 Request access
- 3 Prove identity
- 4 Get radio resources
- 5 Route voice or data
- 6 Move without disappearing

The node names change. The jobs are surprisingly persistent.

1G connected radio voice to the telephone network



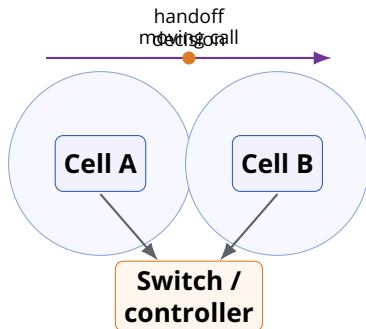
A circuit stays reserved while the call is active.

The path was a circuit.

- Analog radio in the cell
- MTSO coordinated towers and calls
- PSTN carried the call onward

A mobile call was still a telephone call.

Mobility means changing the network while using it

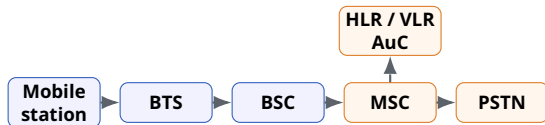


A moving car creates three questions:

- Who measures the signal?
- Who chooses the next cell?
- Who keeps the call alive?

What happens if handoff is late? What if it is too early?

2G split the base station into radio and control



BTS talks over radio

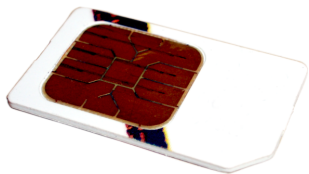
BSC coordinates cells

MSC switches calls

HLR/VLR tracks subscribers

Digital radio made identity and roaming scalable.

The SIM became a network passport



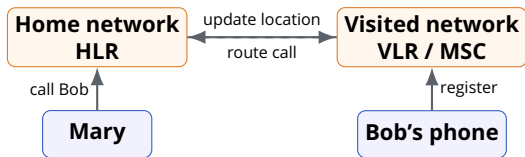
GSM SIM card

Home network knows your subscription, number, and credentials.

Visited network knows where to reach you right now.

If you replace the phone but keep the SIM, what changed in the network?

A call can find you even when you leave home



Bob visits another city.

- 1 Bob registers there.
- 2 The home record is updated.
- 3 Mary's call is forwarded.

Mobility is a database problem as well as a radio problem.

GPRS was an Internet sidecar



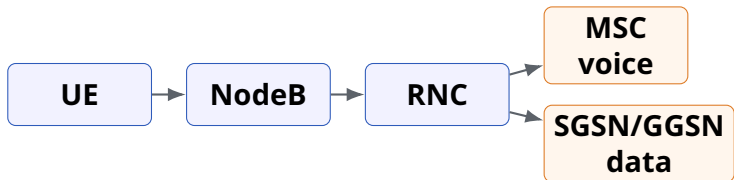
Voice still used the MSC path.

Packet data gained SGSN and GGSN.

The phone now had a persistent route toward the Internet.

2.5G added packet data without replacing the old core.

3G changed the radio faster than the architecture



- UMTS/WCDMA raised mobile-data capability.
- NodeB and RNC echoed the BTS/BSC split.
- Circuit voice and packet data still lived side by side.

[3GPP 3G overview](#)

Voice

Circuit switched

MSC and the telephone world

Data

Packet switched

SGSN/GGSN and the Internet

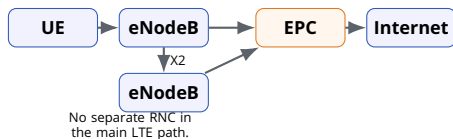
The transition worked. It was not simple.

Can you match the tower name to the generation?

- | | |
|----------|-----------|
| ① BTS | • 2G GSM |
| ② NodeB | • 3G UMTS |
| ③ eNodeB | • 4G LTE |
| ④ gNodeB | • 5G NR |

Which generation removed the separate RNC?

4G flattened the RAN

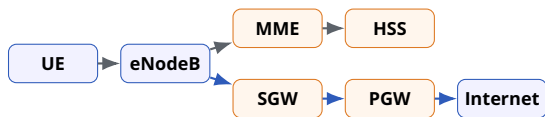


eNodeB absorbed radio control that had lived in the RNC.

Neighboring eNodeBs could coordinate over X2.

The core became the Evolved Packet Core.

The EPC turned the core into packet plumbing



MME mobility and access

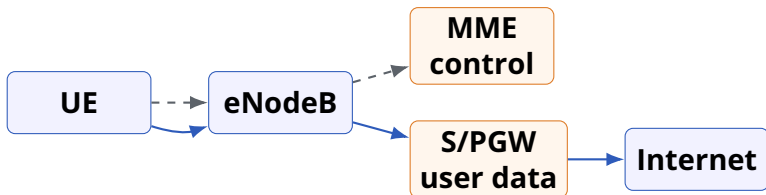
HSS subscriber data

SGW mobility anchor

PGW gateway to Internet

No memorization
test. Learn the jobs.

Control asks; the user plane moves packets



Control plane

attach, authenticate, move,
establish bearer

User plane

carry the voice, video, game, or
web packets

In 4G, a phone call had to become an Internet app

Early LTE

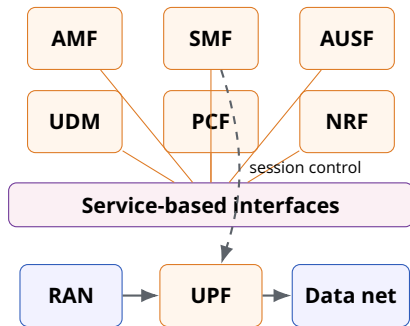
Data on LTE
Voice may fall back to 2G/3G

VoLTE

Voice carried as managed IP
service
IMS provides call control

Why was “all-IP” a bigger change than simply making the radio faster?

5G made the core look more like a cloud platform



Nokia 5G Core Service Based Architecture

AMF access + mobility

SMF session manager

UPF packet forwarding

AUSF/UDM identity

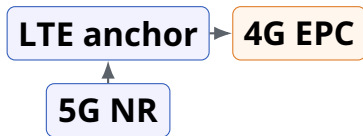
PCF policy

NRF service directory

Functions discover
and consume services.

A 5G radio does not guarantee a 5G core

NSA



Fast deployment; inherits parts of 4G architecture.

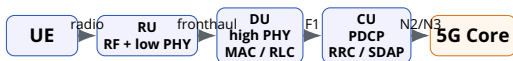
AT&T: "Not all 5G is created equal," 2025

SA



Enables the full 5G system architecture.

The 5G base station can be split across locations



A gNodeB can be a system spread across several physical places.

RU RF + lower PHY

DU upper PHY, MAC, RLC

CU PDCP, RRC, SDAP

Now the question is: who owns each box and each interface?

Can base-station security become an app?

Who installs it? Who approves it? What happens at 2:00 a.m.?

New defense



Vendor ticket



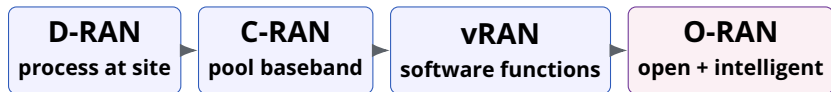
New image



Maintenance window

A programmable RAN asks: can a network behavior become a tested software application?

Four RAN models, four different questions



Virtualization

What runs where?

Disaggregation

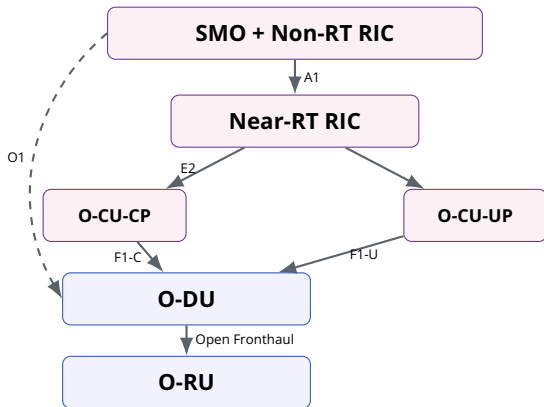
Which functions split?

Open interfaces

Who can connect?

Cloud software does not automatically create an open, multi-vendor RAN.

O-RAN opens the seams—and adds control loops



O-RAN ALLIANCE architecture, redrawn from 2023 white paper

RAN chain

O-CU-CP / O-CU-UP
O-DU / O-RU

Control

SMO + Non-RT RIC
Near-RT RIC

Open seams

A1, E2, O1,
Open Fronthaul

Open RAN is not the same as open-source RAN

Open RAN

- Published interfaces
- Disaggregated functions
- Multi-vendor is possible

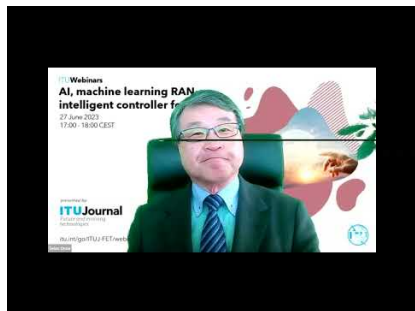
Open source

- Source code is available
- License permits reuse
- Interface may still be proprietary

If two boxes claim the same interface, will they work when plugged together?

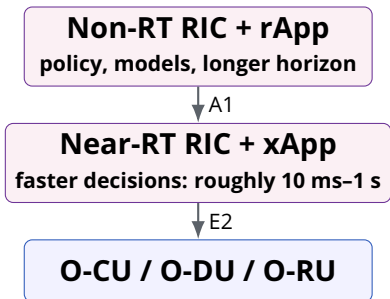
A specification begins interoperability; profiles, testing, and integration finish it.

RIC makes optimization an application



[Play: a real RIC demo \(0:18–1:12\)](#)

ITU, O-RAN / RIC demonstration



Example: traffic steering
rApp sets policy; xApp reacts to load;
RAN executes.

Opening the RAN creates both options and work

Options

- More vendor choice
- Faster software experiments
- Programmable optimization

Work

- Fronthaul and timing
- Integration and testing
- Larger security surface

Would you rather debug one black box or five open interfaces?

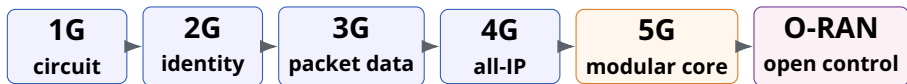
Openness moves work. It does not make the work disappear.

Five generations in one low-resolution map

	Radio	Control	Core	Mode	Example
1G	Base station	MTSO	Analog mobile core	Circuit	Voice in a car
2G	BTS	BSC	MSC + identity DBs	Circuit; GPRS packet	SMS, roaming
3G	NodeB	RNC	CS + PS cores	Both	Mobile web / video
4G	eNodeB	In eNodeB	EPC, all-IP	Packet	Maps, apps, streaming
5G	gNodeB	CU / DU / RU	Service-based 5GC	Packet + QoS flows	Flexible services + edge

The radio name changes. The deeper story is where control, state, and forwarding live.

Follow the same action one last time



The packet still needs **radio access**,
identity + policy + routing,
and now sometimes **programmable control**.

Each generation relocates functions; O-RAN changes who may build and control them.

Exit ticket: draw the boundaries



- 1 Mark the **RAN** and **Core** boundary.
- 2 Add one **O-RAN** interface or controller.
- 3 Answer in one sentence: **What changes next—radio, core, or who can control the network?**

Appendix: the acronym decoder ring

Era	Node	Meaning / job
1G	MTSO	Mobile Telephone Switching Office; switching and mobility control
2G	BTS / BSC	Base Transceiver Station / Base Station Controller
2G/3G	MSC, HLR, VLR, AuC	Voice switch; home/visitor state; authentication
GPRS	SGSN / GGSN	Packet mobility + gateway to external packet networks
3G	NodeB / RNC	Radio node / Radio Network Controller
4G	eNodeB	LTE radio node; absorbs most RNC control
4G	MME, HSS, S-GW, P-GW	Mobility; subscriber data; serving and packet gateways
5G	gNodeB	5G radio node, often split into CU / DU / RU
5GC	AMF, SMF, UPF	Access/mobility; session control; packet forwarding
5GC	AUSF, UDM, PCF, NRF	Authentication; subscriber data; policy; service registry
O-RAN	SMO / RIC	Management/orchestration and intelligent control

Appendix: interface map

Uu UE ↔ radio: air

S1 / NG RAN ↔ core

X2 / Xn neighbor handover

F1 O-CU ↔ O-DU

Open FH O-DU ↔ O-RU

E2 Near-RT RIC ↔ RAN

A1 Non-RT RIC ↔ Near-RT RIC

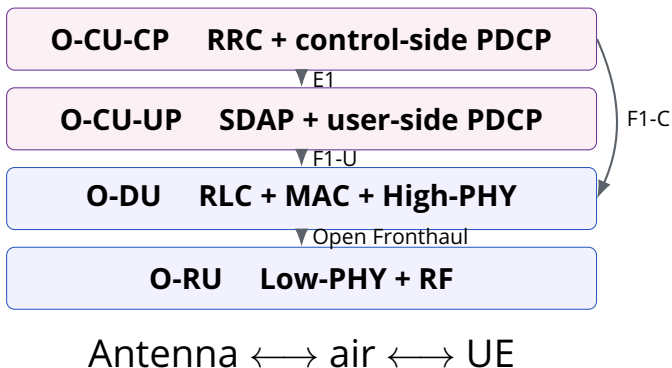
O1 SMO ↔ managed functions

O2 SMO ↔ cloud platform

Interfaces expose coordination;
they do not remove it.

[O-RAN ALLIANCE architecture and resources](#)

Appendix: where the protocol stack is split



A split is a placement decision: latency, bandwidth, compute, and ownership all move with it.

Appendix: handover changed owners, not purpose

Era	Main controller	What crosses the boundary?	What must survive?
1G	MTSO	Analog voice circuit	The conversation
2G	BSC / MSC	Radio channel + circuit state	Call and subscriber context
3G	RNC / core	Radio bearer; CS or PS state	Voice or packet session
4G	eNodeB + EPC	IP bearer and mobility context	All-IP session
5G	gNodeB + 5GC	PDU session, QoS flows, policy	Service continuity
O-RAN	Same 3GPP functions	RIC may influence parameters	Safe control under open APIs

Which is harder to move—a radio channel, a packet tunnel, or its policy?

Sources and further reading

- **3GPP TS 23.501** — authoritative 5G system architecture.
- **Nokia: 5G Core service-based architecture** — readable SBA role map.
- **O-RAN Architecture and Resources** — interfaces, RICs, and current specifications.
- **O-RAN ALLIANCE overview (2026)** — current official overview.
- **AT&T 5G Standalone announcement (2025)** — why a 5G icon can precede 5G Core.
- **Course references:** local materials in `reference/` were used for background only, not reproduced as slide images.

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